Project Report Format

1. INTRODUCTION

1.1 Project Overview

Health360 is a healthcare data analysis project developed using Tableau. It focuses on analyzing healthcare information such as diabetes, heart disease, cholesterol, blood pressure, age, and gender. The project uses data visualization to present important healthcare patterns through an interactive dashboard containing nine visualizations. It helps users understand the dataset and explore relationships between different health indicators.

1.2 Purpose

The main purpose of Health360 is to analyze healthcare data and identify useful patterns related to common health conditions. It aims to compare diabetes and heart disease records, understand cholesterol and blood pressure trends, and study patient distribution by age and gender. The Tableau dashboard makes complex data easier to understand and supports better data-driven insights.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare data contains important information about patients, but understanding patterns across different health conditions can be difficult when the data is presented in raw form. It is challenging to compare diabetes, heart disease, cholesterol, blood pressure, age, and gender without proper visualization. Therefore, this project aims to organize and analyze healthcare data using Tableau dashboards to make these patterns easier to understand.

2.2 Empathy Map Canvas

The project considers the needs of people who want to understand healthcare information easily. Users need clear and simple visualizations to compare health conditions and identify patterns

They may face difficulty understanding large amounts of raw data. Health360 addresses this need by presenting healthcare information through easy-to-understand Tableau charts and dashboards.

2.3 Brainstorming

During the brainstorming phase, different ideas were considered to make healthcare data easier to understand. Tableau was selected as the visualization tool because it provides interactive charts and dashboards. The project focuses on nine visualizations covering diabetes, heart disease, cholesterol, blood pressure, age, and gender. These visualizations help users explore healthcare patterns in a simple and organized way.

3. **REQUIREMENT ANALYSIS**

3.1 Customer Journey map

Users open the Health360 dashboard, explore healthcare charts, compare health conditions, and understand important patterns through visualizations.

3.2 Solution Requirement

The project requires a healthcare dataset, Tableau software, and an interactive dashboard with charts to analyze patient health information.

3.3 Data Flow Diagram

The healthcare dataset is imported into Tableau, processed, and converted into nine visualizations. These charts are combined into a dashboard to present useful insights.

3.4 Technology Stack

- Tool: Tableau Public
- Dataset: Smart Healthcare Dataset (CSV)
- Platform: Tableau

- Repository: GitHub
- Output: Interactive Health360 Dashboard

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Health360 dashboard solves the problem by converting healthcare data into simple and interactive visualizations.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard with nine charts to analyze diabetes, heart disease, age, cholesterol, blood pressure, and gender.

4.3 Solution Architecture

The healthcare dataset is loaded into Tableau, processed into visualizations, and combined into one interactive dashboard.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned in stages, including data collection, analysis, dashboard creation, testing, and documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was tested to check whether all nine visualizations displayed correctly. The charts were reviewed for clarity, accuracy, and easy understanding.

7. RESULTS

7.1 Output Screenshots

The final output is an interactive Tableau dashboard containing nine healthcare visualizations. Screenshots of the dashboard are included as project results.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Easy data visualization.

- Helps identify healthcare patterns.

- Simple and interactive dashboard.

Disadvantages:

- Depends on dataset quality.

- Does not provide medical diagnosis.

- Limited to available data.

9. CONCLUSION

Health360 successfully presents healthcare data through nine Tableau visualizations. It makes healthcare patterns easier to explore and improves understanding of data analysis.

10. FUTURE SCOPE

The project can be improved by adding more healthcare datasets, advanced filters, real-time data, and additional visualizations.

11. APPENDIX

Source Code(if any)

Source code is available in the GitHub repository.

Dataset Link

https://drive.google.com/file/d/1IWWhqlPLfei01ebJI2zWSddt_7yJu-Mg/view?usp=drive_link

GitHub & Project Demo Link

Projectb demo link:
https://drive.google.com/file/d/1ZYsZ3o_J8ND87MfCjecUBoMlKsobhH_E/view?usp=drive_link

Git hub :
https://github.com/en25cs3011022-ai/Health360-Healthcare-analysis